

Scalability & Future Plan

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the A Comprehensive Measure of Well-being project can be enhanced to support more users, larger datasets, and advanced functionalities. It also presents a roadmap for future improvements to increase the usability, scalability, and overall impact of the HDI prediction system.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application runs only on a local Flask server.	Users cannot access the application remotely.	High
2	The model is trained on a static HDI dataset.	Predictions may not reflect newly available data.	High
3	Only Linear Regression is used for prediction.	Prediction accuracy may be improved using advanced algorithms.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users only.	Deploy the application on cloud platforms such as Render, Azure, or AWS to support multiple concurrent users.
2	Data Storage	Uses a CSV file and a Pickle model.	Integrate a relational database such as MySQL or PostgreSQL for efficient data storage and management.
3	Performance	Single-machine execution.	Optimize the model, implement caching, and use a production WSGI server (e.g., Gunicorn) for improved performance.
4	Security	No authentication or user management.	Add user authentication, input validation, HTTPS, and role-based access control to improve security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform for public access.	1 Month	Enables users to access the application online.
Phase 2	Integrate an SQL database to store prediction history and user records.	2 Months	Improves data management and scalability.
Phase 3	Implement advanced Machine Learning models	3 Months	Enhances prediction accuracy and reliability.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
	(Random Forest, XGBoost, etc.) and compare performance.		
Phase 4	Add interactive dashboards, data visualizations, downloadable reports, and user authentication.	4-6 Months	Improves user experience and expands the application's functionality.